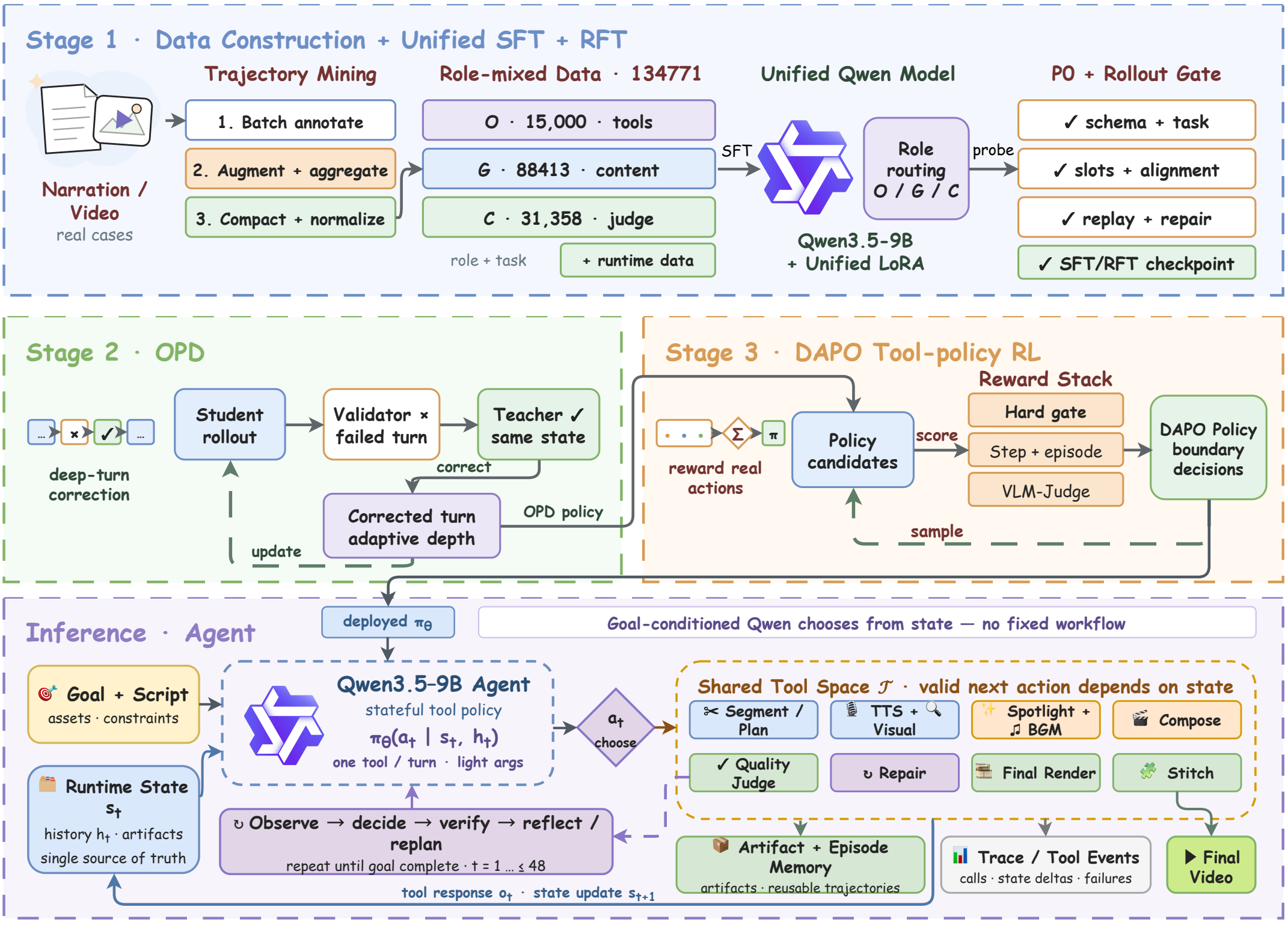

Stage 1 · Data Construction + Unified SFT + RFT
Trajectory Mining
Role-mixed Data · 134771
Unified Qwen Model
PO + Rollout Gate
Narration / Video
real cases
1. Batch annotate
2. Augment + aggregate
3. Compact + normalize
O · 15,000 · tools
G · 88413 · content
C · 31,358 · judge
role + task
+ runtime data
SFT
Qwen3.5-9B
+ Unified LoRA
Role routing
O / G / C
probe
✓ schema + task
✓ slots + alignment
✓ replay + repair
✓ SFT/RFT checkpoint
Stage 2 · OPD
deep-turn correction
Student rollout
Validator × failed turn
Teacher ✓ same state
correct
Corrected turn adaptive depth
OPD policy
update
Stage 3 · DAPO Tool-policy RL
Reward Stack
reward real actions
Policy candidates
score
Hard gate
Step + episode
VLM-Judge
DAPO Policy boundary decisions
sample
Inference · Agent
deployed π_θ
Goal-conditioned Qwen chooses from state — no fixed workflow
Goal + Script
assets · constraints
Qwen3.5-9B Agent
stateful tool policy
$\pi_\theta(a_t \mid s_t, h_t)$
one tool / turn · light args
a_t choose
Shared Tool Space $\mathcal{T}$ · valid next action depends on state
Segment / Plan
TTS + Visual
Spotlight + BGM
Compose
Quality Judge
Repair
Final Render
Stitch
Runtime State s_t
history h_t · artifacts
single source of truth
Observe → decide → verify → reflect / replan
repeat until goal complete · t = 1 … ≤ 48
Artifact + Episode Memory
artifacts · reusable trajectories
Trace / Tool Events
calls · state deltas · failures
Final Video
tool response o_t · state update s_{t+1}